FRAMEWORK · VERSION 1.0 · UPDATED 12 OCTOBER 2026

MAvHU MRV Taxonomy

The classification framework behind MAvHU's MRV infrastructure — every element of the evidence chain, from the capture of a plot boundary to the disclosure of a verified outcome, mapped to the standards it serves.

The taxonomy exists for three reasons. It is the **product architecture**: a shared vocabulary for what MAvHU does. It is the **standards crosswalk**: the evidence that MAvHU's methods map to Verra VCS, Gold Standard, the ICVCM Core Carbon Principles, Article 6 of the Paris Agreement, the Enhanced Transparency Framework and EUDR. And it is the **engagement index**: the structure a stakeholder walks through in a demonstration.

> This taxonomy is the reference for the MRV Infrastructure Profile: where the profile makes a claim, the taxonomy identifies the element behind it.

> One structure, three uses. Adding a framework is a new column in the mapping, not a rebuild of the platform.

Nine domains cover the full evidence chain. Each element carries a code, a description, the MAvHU layer it belongs to — Sky, Earth, AI or API — and a status.

> **Status.** *Live*: in production. *Designed*: built to the requirements of the relevant standard, not yet live or certified. *Planned*: on the roadmap.

The nine domains

1 Data Capture

Everything that brings a field, a plot or a record into the evidence chain.

Code	Element	Description	Layer	Status
1.1.1	**GPS polygons**	Plot boundary captured as a geolocated polygon.	Sky / Earth	**Live**
1.1.2	**Forest boundaries**	Stand and forest boundaries for carbon and land-use projects.	Sky	**Live**
1.1.3	**Plot registration**	Plot registered once and referenced thereafter.	AI	**Live**
1.2.1	**Farmer consent records**	Consent captured from field teams, partners and existing registries.	Earth	**Live**
1.2.2	**Data source provenance**	Every record traceable to its origin.	AI	**Live**
1.2.3	**Chain of custody**	Unbroken custody from capture to disclosure.	AI	**Live**
1.3.1	**Manual entry**	Field-team manual entry.	Earth	**Live**
1.3.2	**Bulk upload**	Bulk upload of existing records.	API	**Live**
1.3.3	**API ingestion**	Programmatic ingestion from partner systems.	API	**Live**
1.3.4	**Registry data import**	Import from existing registries.	API	**Live**

2 Earth Observation

The Sky layer: satellite and reanalysis evidence.

Code	Element	Description	Layer	Status
2.1.1	**Optical imagery**	Satellite optical imagery for land-cover assessment.	Sky	**Live**
2.1.2	**Radar**	Radar imagery for cloud-penetrating observation.	Sky	**Designed**
2.1.3	**Reanalysis datasets**	Reanalysis data already in operational use.	Sky	**Live**
2.2	**Drone validation layer**	Drone imagery validating satellite evidence.	Sky	**Designed**
2.3.1	**Deforestation detection**	Clearing flagged against the regulatory cut-off.	Sky	**Live**
2.3.2	**Degradation detection**	Degradation flagged before it becomes a credit event.	Sky	**Live**
2.3.3	**Overlap detection**	Overlapping claims on the same land flagged.	Sky	**Live**
2.3.4	**Regulatory cut-off check**	Land-cover change checked against 31 December 2020.	Sky	**Live**

3 Ground-truthing

The Earth layer: field verification where satellite evidence is uncertain.

Code	Element	Description	Layer	Status
3.1	**Targeted field verification**	Field checks where satellite evidence is uncertain or risk is high.	Earth	**Live**
3.2	**Sampling design**	Sample-based checks of registers and programme areas.	Earth	**Live**
3.3	**Bayesian optimisation**	Bayesian optimisation directing where to ground-truth.	Earth	**Designed**
3.4	**Field team data capture**	Structured capture by field teams.	Earth	**Live**

4 Quantification

Turning evidence into measured indicators.

Code	Element	Description	Layer	Status
4.1	**Carbon & GHG indicators**	Carbon and GHG quantified under recognised standards.	AI	**Live**
4.2	**Land-health indicators**	Land-health indicators.	AI	**Live**
4.3	**Soil indicators**	Soil indicators from the estate pilot.	AI	**Live**
4.4	**Crop indicators**	Crop indicators from the estate pilot.	AI	**Live**
4.5	**Baselines**	Baseline evidence for project design.	AI	**Designed**
4.6	**Additionality**	Additionality evidence for project design.	AI	**Designed**
4.7	**Permanence & reversal**	Permanence and reversal monitoring.	AI	**Designed**
4.8	**Leakage**	Leakage checks.	AI	**Designed**

5 Verification & Assurance

The rail under every step of the evidence chain.

Code	Element	Description	Layer	Status
5.1	**Uncertainty quantification**	Uncertainty quantified at every stage and combined per output.	AI	**Designed**
5.2	**Conservative deduction**	Conservative deduction where uncertainty cannot be quantified.	AI	**Designed**
5.3	**QA/QC & cross-checks**	Cross-checks against other data sources.	AI	**Designed**
5.4	**Human review & approval**	Statistical flags route to human reviewers; nothing auto-published.	AI	**Live**
5.5	**Failure & backup protocol**	Backup data plan when monitoring fails.	AI	**Designed**
5.6	**VVB verification**	Independent VVBs accredited to ISO 14065 verify.	External (VVB)	***Planned***
5.7	**Algorithm verification**	The calculation algorithm verified, not only its outputs.	AI	***Planned***

6 Evidence & Audit

The persistent, exportable evidence base.

Code	Element	Description	Layer	Status
6.1	**Tamper-evident audit trail**	Locked, tamper-proof audit trail with full export.	AI	**Live**
6.2	**Evidence export**	Audit-ready evidence exported for due diligence.	API	**Live**
6.3	**Data ownership**	Full ownership and export-ready data at any time.	All layers	**Live**
6.4	**Retention & archiving**	Persistent trail replacing data rebuilt each cycle.	AI	**Live**

7 Reporting & Disclosure

Where the evidence goes.

Code	Element	Description	Layer	Status
7.1	**EUDR due diligence**	Plot-level GPS, deforestation-free, legality, due diligence statement.	API	**Live**
7.2	**CBAM**	Carbon border adjustment evidence.	API	**Designed**
7.3	**Scope 3 & ESG**	Scope 3 and ESG evidence on verified plot data.	API	**Live**
7.4	**BTR & GHG inventory**	Mitigation and land-sector inputs to a Biennial Transparency Report.	API	**Designed**
7.5	**NDC tracking**	Indicators used to track progress against an NDC.	API	**Designed**
7.6	**Article 6 reporting**	Authorisation, first transfer, corresponding adjustments.	API	**Designed**
7.7	**Credit issuance evidence**	Evidence supporting credit issuance.	API	**Designed**

8 Governance & Integrity

How the platform itself is governed.

Code	Element	Description	Layer	Status
8.1	**Tool approval & re-approval**	Documented approval before deployment and re-approval triggers.	All layers	**Designed**
8.2	**Incident handling**	Defined incident detection, remedy and disclosure.	All layers	**Designed**
8.3	**Conflict of interest**	Policy separating tool provision from verification.	All layers	**Designed**
8.4	**Cybersecurity & data integrity**	Designed to ISO/IEC 27001 controls (certification in progress); tamper-evident logging.	All layers	**Designed**
8.5	**KYC/AML at registration**	Know-your-counterparty checks at registration.	All layers	**Designed**

9 Interoperability & Registry

How MAvHU data moves into other systems.

Code	Element	Description	Layer	Status
9.1	**API delivery**	Delivery into client systems by API.	API	**Live**
9.2	**Common data model**	Common data model, identifiers and metadata.	API	**Designed**
9.3	**Unique unit identifiers**	Every unit carries a unique identifier.	API	**Designed**
9.4	**National registry integration**	Structured to feed national carbon registries.	API	**Designed**
9.5	**Article 6 registry**	Support for Article 6.2 and 6.4 registry requirements.	API	**Designed**

Standards mapping

Where each element supports evidence for a framework's requirements. A dash means the element is not mapped to a requirement under that framework — not that MAvHU does not provide it.

> **Not an endorsement.** The mapping shows where an element supports a framework's requirements. It is not endorsement, approval or accreditation by any standards body.

Element	Verra VCS	Gold Standard	ICVCM CCP	Article 6	ETF / BTR	EUDR
GPS polygons	✓	✓	✓	✓	✓	✓
Farmer consent records	✓	✓	✓	—	—	✓
Chain of custody	✓	✓	✓	✓	✓	✓
Optical imagery	✓	✓	✓	✓	✓	✓
Radar	✓	✓	✓	—	—	✓
Reanalysis datasets	✓	✓	✓	—	✓	—
Drone validation layer	✓	✓	✓	—	—	—
Deforestation detection	✓	✓	✓	—	✓	✓
Degradation detection	✓	✓	✓	✓	✓	—
Overlap detection	✓	✓	✓	✓	—	—
Regulatory cut-off check	—	—	—	—	—	✓
Targeted field verification	✓	✓	✓	✓	✓	✓
Bayesian optimisation	—	—	—	—	—	—
Carbon & GHG indicators	✓	✓	✓	✓	✓	—

Element	Verra VCS	Gold Standard	ICVCM CCP	Article 6	ETF / BTR	EUDR
Land-health indicators	✓	✓	—	—	✓	—
Baselines	✓	✓	✓	✓	—	—
Additionality	✓	✓	✓	✓	—	—
Permanence & reversal	✓	✓	✓	✓	—	—
Leakage	✓	✓	✓	✓	—	—
Uncertainty quantification	✓	✓	✓	✓	✓	—
Conservative deduction	✓	✓	✓	✓	—	—
QA/QC & cross-checks	✓	✓	✓	✓	✓	—
Human review & approval	✓	✓	✓	—	—	—
Failure & backup protocol	—	✓	—	—	—	—
VVB verification	✓	✓	✓	✓	—	—
Algorithm verification	—	✓	—	—	—	—
Tamper-evident audit trail	✓	✓	✓	✓	✓	✓
Data ownership	—	—	—	—	—	—
EUDR due diligence	—	—	—	—	—	✓
CBAM	—	—	—	—	—	—
Scope 3 & ESG	—	—	—	—	—	—
BTR & GHG inventory	—	—	—	—	✓	—
NDC tracking	—	—	—	—	✓	—
Article 6 reporting	—	—	—	✓	—	—
Credit issuance evidence	✓	✓	✓	✓	—	—
Tool approval & re-approval	—	—	✓	—	—	—
Incident handling	—	—	✓	—	—	—
Conflict of interest	—	—	✓	—	—	—
Cybersecurity & data integrity	—	—	✓	—	—	—
KYC/AML at registration	—	—	✓	—	—	—
API delivery	—	—	—	—	—	—
Common data model	—	—	✓	—	—	—
Unique unit identifiers	—	—	—	✓	—	—
National registry integration	—	—	—	✓	—	—
Article 6 registry	—	—	—	✓	—	—

Version history

- **1.0** — 12 October 2026. Initial publication: nine domains, status per element, standards mapping.

MAvHU Africa — Africa's Climate Data Infrastructure, built where the data is needed.

Version 1.0 · Last updated 12 October 2026. A living document: elements are added as the platform extends, and the standards mapping is updated as frameworks change.

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